

BROUGHTON ISLAND, NU, Canada

WMO: 710960

Lat: 67.536N

Lon: 63.789W

Elev: 584

StdP: 94.50

Time Zone: -5.00 (NAE)

Period: 82-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.6	-33.8	-39.2	0.1	-35.7	-38.3	0.1	-34.9	20.6	-20.3	18.3	-23.3	3.1	0	1.061

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	14.1	9.7	12.3	8.1	10.7	6.9	9.2	12.1	7.3	10.3	6.2	9.1	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.8	10.7	5.1	5.8	8.3	3.8	5.3	7.2	28.8	12.3	24.6	10.3	22.1	9.2	15.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.1	11.9	DB	-35.7	16.0	2.5	1.9	-37.5	17.4	-38.9	18.5	-40.3	19.6	-42.1	21.0	(4)
(5)			WB	-36.6	10.5	1.7	2.0	-37.8	12.0	-38.8	13.2	-39.7	14.3	-41.0	15.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.6	-24.4	-25.3	-22.6	-16.2	-7.8	0.6	5.7	4.2	-1.7	-6.9	-13.5	-20.3	(6)	
	DBStd	11.81	5.46	5.71	6.08	5.66	4.73	3.90	3.99	3.84	3.17	3.52	5.05	5.74	(7)	
	HDD10.0	7528	1065	988	1011	784	551	283	143	183	350	524	704	941	(8)	
	HDD18.3	10554	1323	1222	1269	1034	809	533	391	437	600	782	954	1199	(9)	
	CDD10.0	15	0	0	0	0	0	1	11	4	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	4.4	5.0	4.7	4.5	4.1	4.1	4.1	3.7	4.0	4.4	4.5	4.8	4.7	(14)	
(15) Precipitation	PrecAvg	289	9	9	5	14	34	28	23	29	41	52	35	11	(15)	
	PrecMax	460	73	37	24	50	134	112	57	68	108	165	78	39	(16)	
	PrecMin	180	0	0	0	0	0	0	1	8	10	6	2	0	(17)	
	PrecStd	73	14	10	5	11	28	27	17	17	24	31	21	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.9	-4.8	-2.9	0.3	7.3	13.3	16.8	15.4	9.3	4.8	0.0	-3.3	(19)	
		MCWB	-5.5	-5.1	-3.9	-1.6	4.2	6.5	14.8	9.1	6.2	4.6	-0.1	-3.6	(20)	
	2%	DB	-10.5	-9.8	-7.9	-3.1	3.4	10.2	14.9	13.2	6.7	1.2	-3.3	-6.9	(21)	
		MCWB	-10.9	-10.3	-8.5	-3.5	1.6	6.5	10.9	7.5	4.3	0.1	-3.7	-7.4	(22)	
	5%	DB	-14.3	-14.8	-11.3	-5.6	0.8	8.1	13.3	11.5	4.8	-1.0	-5.0	-9.5	(23)	
		MCWB	-14.7	-15.2	-11.8	-6.0	0.1	4.9	9.5	6.7	3.0	-1.7	-5.5	-9.9	(24)	
	10%	DB	-17.0	-18.2	-14.1	-8.2	-1.4	6.3	11.7	9.9	3.0	-2.5	-6.7	-12.5	(25)	
		MCWB	-17.2	-18.5	-14.5	-8.6	-1.7	3.6	7.7	6.3	1.8	-2.9	-7.2	-12.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.3	-4.1	-5.5	-2.6	2.7	7.6	14.1	9.8	8.3	4.8	-0.7	-7.4	(27)	
		MCDB	-9.8	-4.0	-4.7	-2.3	3.4	10.2	15.3	13.9	11.5	6.0	-0.7	-7.1	(28)	
	2%	WB	-14.1	-12.7	-11.1	-4.7	1.0	6.3	10.6	7.1	4.8	-0.3	-5.0	-9.6	(29)	
		MCDB	-13.9	-12.5	-10.8	-4.3	2.2	9.1	12.5	10.4	6.5	0.2	-4.7	-9.1	(30)	
	5%	WB	-16.4	-16.7	-14.2	-7.5	-0.8	5.0	8.7	6.1	2.4	-2.0	-6.4	-12.5	(31)	
		MCDB	-16.1	-16.4	-13.9	-7.1	-0.2	7.2	11.7	8.9	3.7	-1.7	-5.8	-12.1	(32)	
	10%	WB	-18.9	-19.8	-17.5	-10.0	-2.5	3.7	7.1	5.1	1.0	-3.7	-8.7	-15.5	(33)	
		MCDB	-18.6	-19.5	-17.2	-9.6	-2.1	6.0	10.2	7.4	2.0	-3.3	-8.3	-15.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.4	4.9	4.8	4.0	4.0	4.6	4.1	2.8	3.0	3.5	4.0	(35)	
		MCDBR	6.4	7.4	7.1	7.3	6.4	6.1	5.7	5.7	5.3	4.8	5.0	6.0	(36)	
	5% WB	MCWBR	6.0	6.4	8.4	7.9	4.8	4.2	3.7	4.4	3.7	5.0	5.1	5.6	(37)	
		MCWBR	5.3	6.4	8.1	6.7	5.1	5.4	5.1	5.4	4.3	5.0	5.1	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.146	0.192	0.233	0.277	0.307	0.302	0.313	0.311	0.272	0.227	0.153	N/A	(40)	
		taud	1.861	2.096	2.176	2.077	2.072	2.250	2.400	2.469	2.463	2.203	1.928	N/A	(41)	
	Ebn at Noon		200	677	835	873	879	899	880	844	796	602	176	0	(42)	
		Edh at Noon	10	46	78	116	132	114	96	80	61	41	9	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.05	0.64	2.38	4.81	6.62	6.86	5.53	3.62	2.04	0.76	0.10	0.00	(44)	
	RadStd		0.00	0.04	0.10	0.22	0.24	0.47	0.40	0.22	0.14	0.05	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.68	+1.50	N/A	+0.51	N/A	N/A	-252	-261	+2	N/A
(47) Regional (5 neighbors)		+0.62	+1.46	+1.78	N/A	N/A	N/A	-218	-222	N/A	N/A

Nomenclature: See separate page